HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Managed a team of 4 engineers and technicians, delivering technically sound engineering solutions for bridge inspection and maintenance projects.
- Took ownership of projects from initial quotation through to final handover, ensuring quality, cost, and programme criteria were met.
- Prepared and checked designs, calculations, and documentation for structural schemes, aligning with client requirements and technical standards.
- Contributed to the professional development of junior staff, acting as a **Delegated Engineer** for their technical output and quality assurance.
- Liaised with **Transport Infrastructure Ireland (TII)** and **Iarnród Éireann** representatives at peer level, attending progress meetings and resolving technical queries.
- Collaborated on bid preparation and work-winning functions, contributing technical expertise to proposals for infrastructure frameworks.
- Containerized data processing services with **Docker** and deployed them to an **AWS EC2** cluster, wiring up **Jenkins** pipelines for predictable CI/CD.

SKILLS

Engineering Software – LUSAS, Autodesk Structural Bridge Design

Project Management – Quotation Preparation, Scheme Deliverables, Handover

Technical Standards – Client Requirements, CDM, Risk Assessment

Domain Knowledge – Structures, Bridge Inspection, Resilience, Connectivity

Cloud & DevOps – AWS EC2, Docker, Jenkins

Programming Languages – Java, Python, SQL

PROJECTS

Bridge Inspection and Maintenance Platform

- Developed a Java Spring Boot application to manage bridge inspection data, including defect logging and structural integrity assessments.
- Designed REST APIs for integration with existing enterprise systems, facilitating data exchange for maintenance planning and resource allocation.
- Implemented automated reporting features using Python scripts to generate summaries for stakeholders on structural health across a portfolio of bridges.

Structural Analysis Automation Suite

- Engineered Python scripts to automate data extraction and pre-processing for structural analysis software, reducing manual input time by **40%**.
- Developed data validation routines using SQL queries against PostgreSQL to ensure consistency and accuracy of input parameters for analysis models.
- Created interactive dashboards using Power BI to visualize analysis results and key performance indicators for bridge resilience projects.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)